

Tarea 3:

DPS Exercise set 3: Transformada de Fourier

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Objective

Implement the **Discrete Fourier Series (DFS)** for **periodic discrete-time** signals.

You must:

1. Construct **three** periodic discrete signals (one period only).
2. Compute DFS coefficients **manually** using the definition (correlation / inner product).
3. Reconstruct each signal from its coefficients.
4. Compare original vs reconstructed signals.
5. Report reconstruction error.

We work with one period of a discrete-time periodic signal:

- Period:

$$N$$

- Samples:

$$n = 0, 1, \dots, N - 1$$

- Signal values:

$$x[n]$$

Complex exponential basis

$$\phi_k[n] = e^{j\frac{2\pi}{N}kn}, \quad k = 0, 1, \dots, N - 1$$

Analysis (coefficients)

$$X[k] = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}$$

Synthesis (reconstruction)

$$\hat{x}[n] = \sum_{k=0}^{N-1} X[k] e^{j\frac{2\pi}{N}kn}$$

Reconstruction error (RMSE)

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{n=0}^{N-1} |x[n] - \hat{x}[n]|^2}$$

Signal A: Square Wave

$$N_1 = 16$$

$$x_A[n] = 1 \quad 0 \leq n < N_1/2$$

$$x_A[n] = -1 \quad N_1/2 \leq n < N_1$$

Construyendo la señal $N_1/2 = 16/2 = 8$

n	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$x_A[n]$	1	1	1	1	1	1	1	1	-1	-1	-1	-1	-1	-1	-1	-1

Para calcular los coeficientes

$$X[k] = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-j \frac{2\pi}{N} kn}$$

Sustituimos $N = 16$

$$-j \frac{2\pi}{N} kn = -j \frac{2\pi}{16} kn = -j \frac{\pi}{8} kn$$

$$X[k] = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-j \frac{\pi}{8} kn}$$

→ La suma consiste en multiplicar cada valor de la señal por su exponencial y sumar los 16 resultados. Dado que esta señal es $x_A[n] = 1$ para $n = 0, \dots, 7$ y $x_A[n] = -1$ para $n = 8, \dots, 15$, podemos dividir la suma en 2 partes.

$$X[k] = \frac{1}{16} \left[\sum_{n=0}^7 x[n] e^{-j \frac{\pi}{8} kn} - \sum_{n=8}^{15} x[n] e^{-j \frac{\pi}{8} kn} \right]$$

Se agrupan los valores multiplicados por 1

Se agrupan los valores multiplicados por -1

→ Dado que estamos trabajando con periodo par, la primera mitad es exactamente igual a la 2da mitad pero con signo contrario, así que solo basta con sumar los primeros 8 números

→ Haciendo uso de la fórmula de Euler

$$e^{j\theta} = \cos(\theta) + j\sin(\theta)$$

$$e^{-j\theta} = \cos(\theta) - j\sin(\theta)$$

Sustituyendo nuestro problema

$$e^{-j \frac{\pi}{8} kn} = \cos\left(\frac{\pi}{8} kn\right) - j\sin\left(\frac{\pi}{8} kn\right)$$

Parte real

parte imaginaria

$$k = 0 \quad \theta = 0$$

$$X[0] = \frac{1}{16} [1+1+1+1+1+1+1+1-1-1-1-1-1-1-1-1] = 0$$

$$k = 1 \quad \theta = \frac{\pi n}{8}$$

n	cos θ	Sen θ	n	cos θ	Sen θ	n	cos θ	Sen θ
0	1	0	3	0.3827	0.9239	6	-0.7071	0.7071
1	0.9239	0.3827	4	0	1	7	-0.9239	0.3827
2	0.7071	0.7071	5	-0.3827	0.9239			

Sumamos la parte real

$$1 + 0.9234 + 0.7071 + 0.3827 + 0 - 0.3827 - 0.7071 - 0.9239 = 1$$

Como la otra mitad es la misma con signo contrario, sumamos ambas partes

$$1 - (-1) = 2$$

Sumamos la parte imaginaria

$$0 + 0.3827 + 0.7071 + 0.9239 + 1 + 0.9239 + 0.7071 + 0.3827 = 5.0274$$

Sumamos ambas partes

$$5.0274 - (-5.0274) = 10.0548$$

Calculamos el coeficiente

$$X[1] = \frac{1}{16} (2 - j10.0548) = \frac{1}{8} - j0.6259 = 0.6407 \angle -1.3744$$

$$h=2 \quad \theta = \frac{\pi}{8} 2n$$

n	cos θ	sen θ	n	cos θ	sen θ	n	cos θ	sen θ
0	1	0	3	-0.7071	0.7071	6	0	-1
1	0.7071	0.7071	4	-1	0	7	0.7071	-0.7071
2	0	1	5	-0.7071	-0.7071			

Sumamos la parte real

$$1 + 0.7071 + 0 - 0.7071 - 1 - 0.7071 + 0 + 0.7071 = 0$$

Sumamos la parte imaginaria

$$0 + 0.7071 + 1 + 0.7071 + 0 - 0.7071 - 1 - 0.7071 = 0$$

Calculamos el coeficiente

$$X[2] = \frac{1}{16} (0 - j0) = 0$$

$$h=3 \quad \theta = \frac{\pi}{8} 3n$$

n	cos θ	sen θ	n	cos θ	sen θ	n	cos θ	sen θ
0	1	0	3	-0.9238	-0.3827	6	0.7071	0.7071
1	0.3827	0.9139	4	0	-1	7	-0.3827	0.9239
2	-0.7071	0.7071	5	0.9238	-0.3827			

Sumamos la parte real

$$1 + 0.3827 - 0.7071 - 0.9238 + 0 + 0.9238 + 0.7071 - 0.3826 = 1$$

$$1 - (-1) = 2$$

Sumamos la parte imaginaria

$$0 + 0.9139 + 0.7071 - 0.3827 - 1 - 0.3827 + 0.7071 + 0.9239 = 1.4966$$

$$1.4966 - (-1.4966) = 2.9932$$

Calculamos el coeficiente

$$X[3] = \frac{1}{16} (2 - j2.9932) = 0.1250 - j0.1871 = 0.2250 \angle -0.9811$$

$$h=4 \quad \theta = \frac{\pi}{8}n$$

n	cos θ	sen θ	n	cos θ	sen θ	n	cos θ	sen θ
0	1	0	3	0	-1	6	-1	0
1	0	1	4	1	0	7	0	-1
2	-1	0	5	0	1			

Sumamos la parte real = 0

Sumamos la parte imaginaria = 0

$$\text{Calculamos el coeficiente} = X[4] = \frac{1}{16} (0 - j0) = 0$$

$$h=5 \quad \theta = \frac{\pi}{8}n$$

n	cos θ	sen θ	n	cos θ	sen θ	n	cos θ	sen θ
0	1	0	3	0.9239	-0.3827	6	0.7071	-0.7071
1	-0.3827	0.9239	4	0	1	7	0.3827	0.9239
2	-0.7071	-0.7071	5	-0.9239	-0.3827			

Sumamos la parte real = 1

Sumamos la parte imaginaria = 0.6688

$$\text{Calculamos el coeficiente} = X[5] = \frac{1}{16} (2 - j1.3364) = 0.125 - j0.0835 = 0.1503 \angle -0.5890$$

$$h=6 \quad \theta = \frac{\pi}{8}n$$

n	cos θ	sen θ	n	cos θ	sen θ	n	cos θ	sen θ
0	1	0	3	0.7071	0.7071	6	0	1
1	-0.7071	0.7071	4	-1	0	7	-0.7071	-0.7071
2	0	-1	5	0.7071	-0.7071			

Sumamos la parte real = 0

Sumamos la parte imaginaria = 0

$$\text{Calculamos el coeficiente} = X[6] = \frac{1}{16} (0) = 0$$

$$h=7 \quad \theta = \frac{\pi}{8}n$$

n	cos θ	sen θ	n	cos θ	sen θ	n	cos θ	sen θ
0	1	0	3	-0.3827	0.9239	6	-0.7071	-0.7071
1	-0.9239	0.3827	4	0	-1	7	0.9239	0.3827
2	0.7071	-0.7071	5	0.3827	0.9239			

Sumamos la parte real = 1

Sumamos la parte imaginaria = 0.1989

$$\text{Calculamos el coeficiente} = X[7] = \frac{1}{16} (2 - j0.3978) = 0.125 - j0.0249 = 0.1274 \angle -0.1963$$

$$h=8 \quad \theta = \frac{\pi \cdot 8}{8} n$$

n	cos θ	Sen θ	n	cos θ	Sen θ	n	cos θ	Sen θ
0	1	0	3	-1	0	6	1	0
1	-1	0	4	1	0	7	-1	0
2	1	0	5	-1	0			

Sumamos la parte real = 0

Sumamos la parte imaginaria = 0

$$\text{Calculamos el coeficiente} = X[8] = \frac{1}{16} (0 - j0) = 0$$

$$h=9 \quad \theta = \frac{\pi \cdot 9}{8} n$$

n	cos θ	Sen θ	n	cos θ	Sen θ	n	cos θ	Sen θ
0	1	0	3	-0.3827	-0.9239	6	-0.7071	0.7071
1	-0.9239	-0.3827	4	0	1	7	0.9239	-0.3827
2	0.7071	0.7071	5	0.3827	-0.9239			

Sumamos la parte real = 1

Sumamos la parte imaginaria = -0.1989

$$\text{Calculamos el coeficiente} = X[9] = \frac{1}{16} (2 + j0.3978) = 0.125 + j0.0249 = 0.1274 \angle 0.1963$$

$$h=10 \quad \theta = \frac{\pi \cdot 10}{8} n$$

n	cos θ	Sen θ	n	cos θ	Sen θ	n	cos θ	Sen θ
0	1	0	3	0.7071	-0.7071	6	0	-1
1	-0.7071	-0.7071	4	-1	0	7	-0.7071	0.7071
2	0	1	5	0.7071	-0.7071			

Sumamos la parte real = 0

Sumamos la parte imaginaria = 0

$$\text{Calculamos el coeficiente} = X[10] = \frac{1}{16} (0) = 0$$

$$h=11 \quad \theta = \frac{\pi \cdot 11}{8} n$$

n	cos θ	Sen θ	n	cos θ	Sen θ	n	cos θ	Sen θ
0	1	0	3	0.9239	0.3827	6	0.7071	0.7071
1	-0.3827	-0.9239	4	0	-1	7	0.3827	-0.9239
2	-0.7071	0.7071	5	-0.9239	0.3827			

Sumamos la parte real = 1

Sumamos la parte imaginaria = -0.6688

$$\text{Calculamos el coeficiente} = X[11] = \frac{1}{16} (2 + j1.3364) = 0.125 + j0.0835 = 0.1505 \angle 0.5890$$

$$h=12 \quad \theta = \frac{\pi}{8} 12n$$

n	cos θ	sen θ	n	cos θ	sen θ	n	cos θ	sen θ
0	1	0	3	0	1	6	-1	0
1	0	-1	4	1	0	7	0	1
2	-1	0	5	0	-1			

Sumamos la parte real = 0

Sumamos la parte imaginaria = 0

$$\text{Calculamos el coeficiente} = X[] = \frac{1}{16} (0 - j0) = 0$$

$$h=13 \quad \theta = \frac{\pi}{8} 13n$$

n	cos θ	sen θ	n	cos θ	sen θ	n	cos θ	sen θ
0	1	0	3	-0.9239	0.3827	6	0.7071	-0.7071
1	0.3827	-0.9239	4	0	1	7	-0.3827	-0.9239
2	-0.7071	-0.7071	5	0.9239	0.3827			

Sumamos la parte real = 1

Sumamos la parte imaginaria = -1.4956

$$\text{Calculamos el coeficiente} = X[13] = \frac{1}{16} (2 + j2.9932) = 0.125 + j0.1871 = 0.2250 \angle 0.9817$$

$$h=14 \quad \theta = \frac{\pi}{8} 14n$$

n	cos θ	sen θ	n	cos θ	sen θ	n	cos θ	sen θ
0	1	0	3	-0.7071	-0.7071	6	0	1
1	0.7071	-0.7071	4	-1	0	7	0.7071	0.7071
2	0	-1	5	-0.7071	0.7071			

Sumamos la parte real = 0

Sumamos la parte imaginaria = 0

$$\text{Calculamos el coeficiente} = X[14] = \frac{1}{16} (0) = 0$$

$$h=15 \quad \theta = \frac{\pi}{8} 15n$$

n	cos θ	sen θ	n	cos θ	sen θ	n	cos θ	sen θ
0	1	0	3	0.3827	-0.9239	6	-0.7071	-0.7071
1	0.9239	-0.3827	4	0	-1	7	-0.9239	-0.3827
2	0.7071	-0.7071	5	-0.3827	-0.9239			

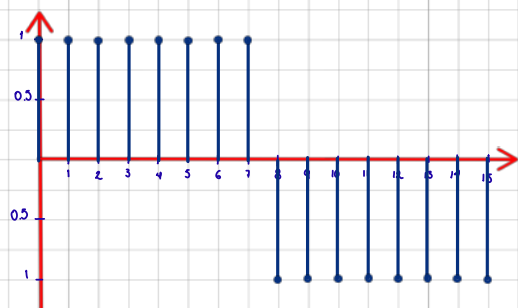
Sumamos la parte real = 1

Sumamos la parte imaginaria = -5.0273

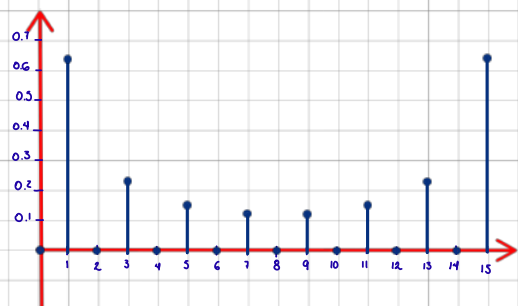
$$\text{Calculamos el coeficiente} = X[15] = \frac{1}{16} (2 + j10.0547) = 0.125 + j0.6284 = 0.6407 \angle 1.3744$$

Graficas

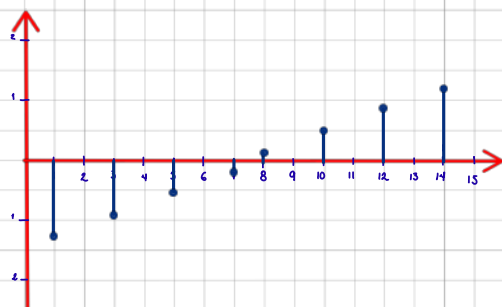
Señal A $x_A[n]$; $N=16$



Grafica de magnitud



Grafica de fase



Signal D : Triangle Wave

$N_x = 32$

Build a discrete triangle over one period, peak amplitud 1 (Or rescale to [-1,1])

Construyendo la señal

n	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$x_a[n]$	-1	-0.875	-0.75	-0.625	-0.5	-0.375	-0.25	-0.125	0	0.125	0.25	0.375	0.5	0.625	0.75	0.875
n	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
	1	0.875	0.75	0.625	0.5	0.375	0.25	0.125	0	-0.125	-0.25	-0.375	-0.5	-0.625	-0.75	-0.875

La primera mitad es igual a segunda mitad con signo contrario

$$X[k] = \frac{1}{32} \left[\sum_{n=0}^{15} x[n] e^{-j\frac{\pi}{16}kn} - \sum_{n=16}^{31} x[n] e^{-j\frac{\pi}{16}kn} \right]$$

Se agrupan los valores multiplicados por 1

Se agrupan los valores multiplicados por -1

Haciendo uso de la formula de Euler

$$e^{j\theta} = \cos(\theta) + j\sin(\theta)$$

$$e^{-j\theta} = \cos(\theta) - j\sin(\theta)$$

Sustituyendo nuestro problema

$$e^{-j\frac{\pi}{16}kn} = \cos\left(\frac{\pi}{16}kn\right) - j\sin\left(\frac{\pi}{16}kn\right)$$

$$h=0 \quad \theta = \frac{\pi}{16} \cdot 0 \cdot n$$

Sumamos la parte real = 0

Sumamos la parte imaginaria = 0

Calculamos el coeficiente = $X[0] = 0$

$$h=1$$

n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$
0	-1	0	4	-0.3536	-0.3536	8	0	0	12	-0.3536	0.3536
1	-0.8582	-0.1707	5	-0.2083	-0.3118	9	-0.0244	0.1226	13	-0.5197	0.3472
2	-0.6929	-0.2870	6	-0.0975	-0.2310	10	-0.0975	0.2310	14	-0.6929	0.2870
3	-0.5197	-0.3472	7	-0.0244	-0.1226	11	-0.2083	0.3118	15	-0.8582	0.1707

Sumamos la parte real = -6.5054

Sumamos la parte imaginaria = 0

Calculamos el coeficiente = $X[1] = \frac{1}{32} (-13.0109 - j0) = -0.4066 - j0 = -0.4066 < \pi$

$h=2$

n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$
0	-1	0	4	0	-0.5	8	0	0	12	0	-0.5
1	-0.8084	-0.3348	5	0.1435	-0.3465	9	-0.1155	-0.0478	13	0.2392	-0.5774
2	-0.5303	-0.5303	6	0.1768	-0.1768	10	-0.1768	-0.1768	14	0.5303	-0.5303
3	-0.2392	-0.5774	7	0.1155	-0.0478	11	-0.1435	-0.3465	15	0.8084	-0.3348

Sumamos la parte real = -1

Sumamos la parte imaginaria = -5.0272

Calculamos el coeficiente = $X[2] = \frac{1}{32} (0 - j0) = 0$

$h=3$

n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$
0	-1	0	4	0.3536	-0.3536	8	0	0	12	0.3536	0.3536
1	-0.7275	-0.4861	5	0.3678	-0.0732	9	0.0694	-0.1039	13	0.1219	0.6130
2	0.2870	-0.6929	6	0.2310	0.0957	10	0.2310	-0.0957	14	0.2870	0.6929
3	0.1219	-0.6130	7	0.0694	0.1039	11	0.3678	0.0732	15	-0.7275	0.4861

Sumamos la parte real = -0.7471

Sumamos la parte imaginaria = 0

Calculamos el coeficiente = $X[3] = \frac{1}{32} (-1.4834 - j0) = -0.0464 - j0 = 0.0464 \angle 3.1416$

$h=4$

n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$
0	-1	0	4	0.5	0	8	0	0	12	-0.5	0
1	-0.6187	-0.6187	5	0.2652	0.2652	9	0.0984	0.0884	13	-0.4419	-0.4419
2	0	-0.75	6	0	0.25	10	0	0.25	14	0	-0.75
3	0.4419	-0.4419	7	-0.0984	0.0884	11	-0.2652	0.2652	15	0.6187	-0.6187

Sumamos la parte real = -1

Sumamos la parte imaginaria = -2.4142

Calculamos el coeficiente = $X[4] = \frac{1}{32} (0 - j0) = 0$

$h=5$

n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$
0	-1	0	4	0.3536	0.3536	8	0	0	12	0.3536	-0.3536
1	-0.4861	-0.7275	5	-0.0732	0.3678	9	-0.1039	0.0694	13	0.6130	0.1219
2	0.2870	-0.6929	6	-0.2310	0.0957	10	-0.2310	-0.0957	14	0.2870	0.6929
3	0.6130	0.1219	7	-0.1039	-0.0694	11	-0.0732	-0.3678	15	-0.4861	0.7275

Sumamos la parte real = -0.2813

Sumamos la parte imaginaria = 0

Calculamos el coeficiente = $X[5] = \frac{1}{32} (-0.5625 - j0) = -0.0176 - j0 = 0.0176 \angle \pi$

$h = 6$

n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$
0	-1	0	4	0	0.5	8	0	0	12	0	0.5
1	-0.3348	-0.8084	5	-0.3465	0.1435	9	-0.0478	-0.1155	13	-0.5774	0.2392
2	0.5303	-0.5303	6	-0.1768	-0.1768	10	0.1768	-0.1768	14	-0.5303	-0.5303
3	0.5774	0.2392	7	0.0478	-0.1155	11	0.3465	0.1435	15	0.3348	-0.8084

Sumamos la parte real = -1

Sumamos la parte imaginaria = -1.4965

Calculamos el coeficiente = $X[6] = \frac{1}{32} (0 - j0) = 0$

$h = 7$

n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$
0	-1	0	4	-0.3536	0.3035	8	0	0	12	-0.3536	-0.3035
1	-0.1707	-0.8582	5	-0.3118	-0.2083	9	0.1226	-0.0244	13	0.3472	-0.5197
2	0.6929	-0.2870	6	0.0957	-0.2310	10	0.0957	0.2310	14	0.6929	0.2870
3	0.3472	0.5197	7	0.1226	0.0244	11	-0.3118	0.2083	15	-0.1707	0.8582

Sumamos la parte real = -0.1553

Sumamos la parte imaginaria = 0

Calculamos el coeficiente = $X[7] = \frac{1}{32} (-0.3106 - j0) = -0.0097 - j0 = 0.0097 < \pi$

$h = 8$

n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$
0	-1	0	4	-0.5	0	8	0	0	12	0.50	0
1	0	-0.8750	5	0	-0.3750	9	0	0.125	13	0	0.625
2	0.75	0	6	0.25	0	10	0.25	0	14	-0.75	0
3	0	0.625	7	0	0.1250	11	0	-0.375	15	0	-0.875

Sumamos la parte real = -1

Sumamos la parte imaginaria = -1

Calculamos el coeficiente = $X[8] = \frac{1}{32} (0 - j0) = 0$

$h = 9$

n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$
0	-1	0	4	-0.3536	-0.3035	8	0	0	12	-0.3536	0.3035
1	0.1707	-0.8582	5	0.3118	-0.2083	9	-0.1226	-0.0244	13	-0.3472	-0.5197
2	0.6929	0.2870	6	0.0957	0.2310	10	0.0957	-0.2310	14	0.6929	-0.2870
3	-0.3472	0.5197	7	-0.1226	0.0244	11	0.3118	0.2083	15	0.1707	0.8582

Sumamos la parte real = -0.1046

Sumamos la parte imaginaria = 0

Calculamos el coeficiente = $X[9] = \frac{1}{32} (-0.2092 - j0) = -0.0065 - j0 = 0.0065 < \pi$

$h=10$

n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$
0	-1	0	4	0	-0.5	8	0	0	12	0	-0.5
1	0.3348	-0.8084	5	0.3465	0.1435	9	0.0478	-0.1155	13	0.5774	0.2392
2	0.5303	0.5303	6	-0.1768	0.1768	10	0.1768	0.1768	14	-0.5303	0.5303
3	-0.5774	0.2392	7	-0.0478	-0.1155	11	-0.3465	0.1435	15	-0.3348	-0.8084

Sumamos la parte real = -1

Sumamos la parte imaginaria = -0.6682

Calculamos el coeficiente = $X[10] = \frac{1}{32} (0 - j0) = 0$

$h=11$

n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$
0	-1	0	4	0.3536	-0.3536	8	0	0	12	0.3536	0.3536
1	0.4861	-0.7275	5	0.0732	0.3678	9	0.1039	0.0694	13	-0.6130	0.1219
2	0.2870	0.6929	6	-0.2310	-0.0957	10	-0.2310	0.0957	14	0.2870	-0.6929
3	-0.6130	-0.1219	7	0.1039	-0.0694	11	0.0732	-0.3678	15	0.4861	0.7275

Sumamos la parte real = -0.0804

Sumamos la parte imaginaria = 0

Calculamos el coeficiente = $X[11] = \frac{1}{32} (-0.1607 - j0) = -0.0050 - j0 = 0.0050 < \pi$

$h=12$

n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$
0	-1	0	4	0.5	0	8	0	0	12	-0.5	0
1	0.6187	-0.6187	5	-0.2652	0.2652	9	-0.0984	0.0884	13	0.4419	-0.4419
2	0	0.75	6	0	-0.25	10	0	-0.25	14	0	0.75
3	-0.4419	-0.4419	7	0.0984	0.0884	11	0.2652	0.2652	15	-0.6187	-0.6187

Sumamos la parte real = -1

Sumamos la parte imaginaria = -0.4142

Calculamos el coeficiente = $X[12] = \frac{1}{32} (0 - j0) = 0$

$h=13$

n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$	n	$x[n] \cos \theta$	$x[n] \sin \theta$
0	-1	0	4	0.3536	0.3536	8	0	0	12	0.3536	-0.3536
1	0.7275	-0.4861	5	-0.3678	-0.0732	9	-0.0694	-0.1039	13	-0.1219	0.6130
2	-0.2870	0.6929	6	0.2310	-0.0957	10	0.2310	0.0957	14	-0.2870	0.6929
3	-0.1219	-0.6130	7	-0.0694	0.1039	11	-0.3678	0.0732	15	0.7275	0.4861

Sumamos la parte real = -0.0683

Sumamos la parte imaginaria = 0

Calculamos el coeficiente = $X[13] = \frac{1}{32} (-0.1365 - j0) = -0.0043 - j0 = 0.0043 < -\pi$

$$h = 14$$

n	x[n] cos θ	x[n] sen θ	n	x[n] cos θ	x[n] sen θ	n	x[n] cos θ	x[n] sen θ	n	x[n] cos θ	x[n] sen θ
0	-1	0	4	0	0.5	8	0	0	12	0	0.5
1	0.8084	-0.3348	5	-0.1435	0.3465	9	0.1155	-0.0478	13	-0.2392	-0.5774
2	-0.5303	0.5303	6	0.1768	0.1768	10	-0.1768	0.1768	14	0.5303	0.5303
3	0.2392	-0.5774	7	-0.1155	-0.0478	11	0.1435	-0.3465	15	-0.8084	-0.3348

$$\text{Sumamos la parte real} = -1$$

$$\text{Sumamos la parte imaginaria} = -0.1989$$

$$\text{Calculamos el coeficiente} = X[14] = \frac{1}{32} (0 - j0) = 0$$

$$h = 15$$

n	x[n] cos θ	x[n] sen θ	n	x[n] cos θ	x[n] sen θ	n	x[n] cos θ	x[n] sen θ	n	x[n] cos θ	x[n] sen θ
0	-1	0	4	-0.3536	0.3536	8	0	0	12	-0.3536	0.3536
1	0.8582	-0.1707	5	0.2083	-0.3118	9	0.0244	0.1226	13	0.5197	0.3472
2	-0.6929	0.2870	6	-0.0975	0.2310	10	-0.0975	-0.2310	14	-0.6929	-0.2870
3	0.5197	-0.3472	7	0.0244	-0.1226	11	0.2083	0.3118	15	0.8582	0.1707

$$\text{Sumamos la parte real} = -0.0631$$

$$\text{Sumamos la parte imaginaria} = 0$$

$$\text{Calculamos el coeficiente} = X[15] = \frac{1}{32} (-0.1262 - j0) = -0.0039 - j0 = 0.0039 \angle -\pi$$

$$h = 16 \quad X[16] = 0$$

$$h = 17 \quad X[17] = 0.00394 \angle \pi$$

$$h = 18 \quad X[18] = 0$$

$$h = 19 \quad X[19] = 0.00427 \angle \pi$$

$$h = 20 \quad X[20] = 0$$

$$h = 21 \quad X[21] = 0.00502 \angle \pi$$

$$h = 22 \quad X[22] = 0$$

$$h = 23 \quad X[23] = 0.00654 \angle \pi$$

$$h = 24 \quad X[24] = 0$$

$$h = 25 \quad X[25] = 0.00971 \angle \pi$$

$$h = 26 \quad X[26] = 0$$

$$h = 27 \quad X[27] = 0.01758 \angle \pi$$

$$h = 28 \quad X[28] = 0$$

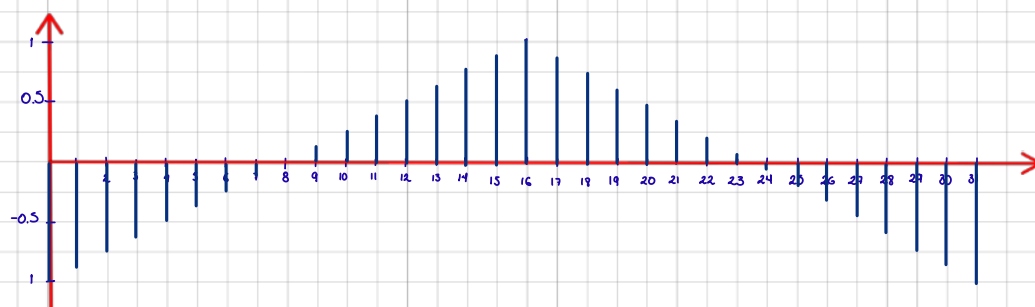
$$h = 29 \quad X[29] = 0.04636 \angle \pi$$

$$h = 30 \quad X[30] = 0$$

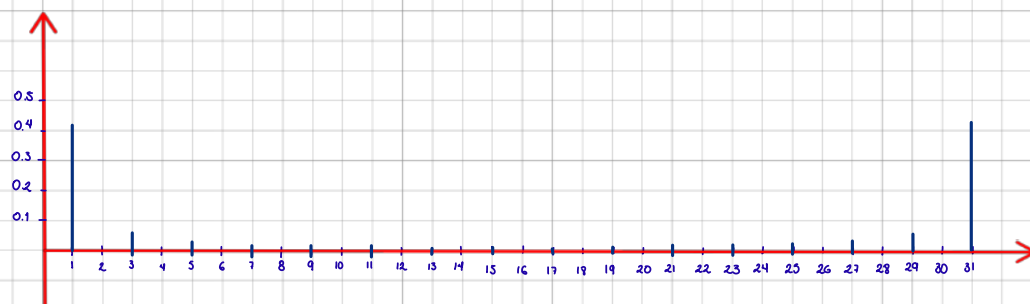
$$h = 31 \quad X[31] = 0.40659 \angle \pi$$

Graficas

Señal



Grafica de magnitud



Grafica de fase



Señal C

La señal es una suma de tres senoidales con frecuencias que son múltiplos enteros de $\omega_0 = \frac{2\pi}{64} = \frac{\pi}{32}$

$$x_c[n] = 0.8 \cos(\omega_0 n) + 0.4 \sin(2\omega_0 n + 0.3) + 0.2 \cos(3\omega_0 n - 0.8)$$

Como la DFT es única, el coeficiente que multiplica a cada $e^{jkh\omega_0 n}$ es directamente $X[k]$. Además, como $e^{j2\pi n} = 1$ una exponencial con frecuencia $-K$ equivale a la frecuencia $64-K$, por lo que los componentes negativos aparecen en $K=63, 62$ y 61 .

Primer término

$$0.8 \cos(\omega_0 n) = 0.4 e^{j\omega_0 n} + 0.4 e^{-j\omega_0 n}$$
$$X[1] = 0.4, \quad X[63] = 0.4$$

Segundo término

$$0.4 \sin(2\omega_0 n + 0.3) = (-0.2j e^{j0.3}) e^{j2\omega_0 n} + 0.2j e^{-j0.3} e^{-j2\omega_0 n}$$
$$X[2] = -0.2j e^{j0.3} = 0.05910 + j0.19107$$
$$X[62] = 0.2j e^{-j0.3} = 0.05910 + j0.19107$$

Tercer término

$$0.2 \cos(3\omega_0 n - 0.8) = (0.1 e^{j0.8}) e^{j3\omega_0 n} + 0.1 e^{-j0.8} e^{-j3\omega_0 n}$$
$$X[3] = 0.1 e^{j0.8} = 0.03967 + j0.07174$$
$$X[61] = 0.1 e^{-j0.8} = 0.03967 + j0.07174$$

Los demás coeficientes valen 0